EECS 113 Final Project Documentation

Steps:

- Attach the GPIO extension board + cobbler board if not done so already.
- 3 pushbuttons, one green LED, one red LED, one blue LED, one PIR sensor, one DHT, and one LCD are needed.
- In Fig. 1 below, we have from left to right the incr. temp button, decr. temp button, and the door/window open/close button. The green LED is hooked to the PIR sensor, the red LED displays if the heat is on, and the blue LED displays if the AC is on.
- Adjusting the PIR screw, R1 as documented in the Freenove tutorial, is necessary so as to have the PIR sensor be responsive to motion immediately and not have up to a 320s delay. Twist the R1 screw counterclockwise to do so.
- Obtain the API key from OpenWeatherMap, the API key will take around 10 minutes to activate.
- Test each component with the python scripts provided by Freenove. Specifically for the DHT, you will have to compile setup.py to obtain the libraries needed to have the DHT running in libdht.so.
- Code will be provided separately, already commented for viewing. Important to have all the separate python files for each component in the same folder so we can import them into our main python script.
- **TO RUN:** download zip file and have everything in one folder, and run the main python file with python3 bms.py. The other files are there for function access for components.
- Note: spacing / placing components was an issue as there is limited space on the breadboard, and the pins on the PIR sensor were too big for the breadboard for some reason, which is why the PIR sensor is in an insecure position.
- Note: PIR sensor not entirely accurate, set to turn off after 10s but shows delay up to 19s, shown in log.txt screenshot below. In the code it is set to turn after 10s.
- Note: Fire alarm set to 90 degrees for testing purposes, in code submitted it is brought back up to the intended 95 degrees.

Final Breadboard Design:

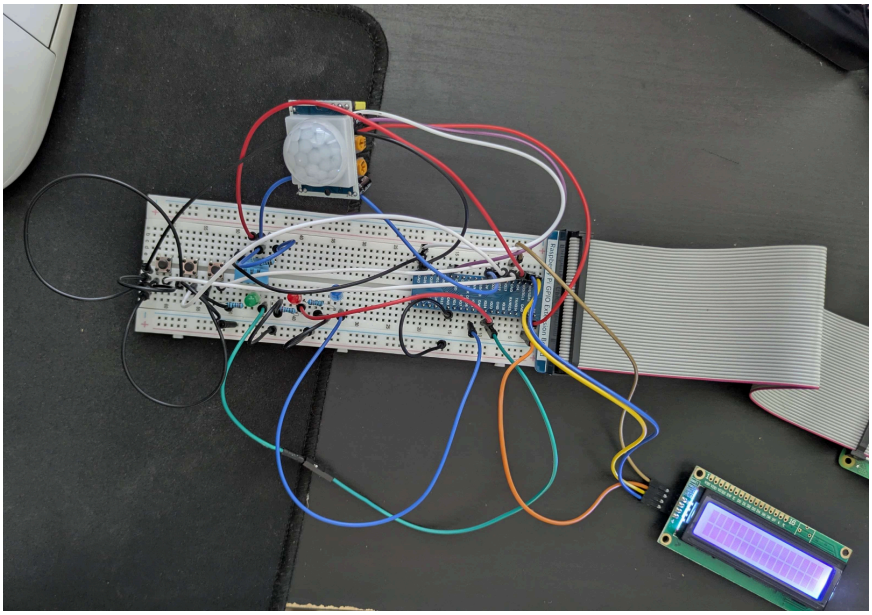

Fig. 1: Final BMS Design on Breadboard

Log.txt Screenshot:

This is the screenshot from the video attached as well, as I forgot to include that in my recording.

bms_project > log.txt

```
11:41:11 HVAC AC
11:41:14 LIGHTS ON
11:41:35 LIGHTS OFF
11:41:43 DOOR OPEN
11:41:44 HVAC OFF
11:41:52 LIGHTS ON
11:41:52 DOOR CLOSED
11:41:54 HVAC AC
11:42:01 TEMP INCREASED TO 73
11:42:05 TEMP DECREASED TO 72
11:42:09 TEMP INCREASED TO 73
11:42:10 TEMP INCREASED TO 74
11:42:10 TEMP INCREASED TO 75
11:42:10 TEMP INCREASED TO 76
11:42:11 TEMP INCREASED TO 77
11:42:11 TEMP INCREASED TO 78
11:42:12 HVAC OFF
11:42:19 LIGHTS OFF
11:42:19 TEMP INCREASED TO 79
11:42:20 TEMP INCREASED TO 80
11:42:20 TEMP INCREASED TO 81
11:42:20 TEMP INCREASED TO 82
11:42:21 TEMP INCREASED TO 83
11:42:21 HVAC HEAT
11:42:24 TEMP INCREASED TO 84
11:42:35 LIGHTS ON
11:42:58 HVAC OFF
11:43:04 DOOR OPEN
11:43:04 FIRE ALARM
11:43:10 LIGHTS OFF
11:43:10 LIGHTS ON
11:43:38 FIRE CLEARED
11:44:07 LIGHTS OFF
```

Pins Used on GPIO:
Inputs for pushbuttons:
17 (incr), 27 (decr), 22 (door / window)

Inputs for LEDS:
Green: 23
Red: 24
Blue: 25

Other Inputs:
dht-11, 4
PIR Sensor, 5
LCD, SDA1 SCl1 (io2 / io3)